Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Water Quality | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Explain why this transect is an estuary rather than a freshwater-only habitat.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: It connects river freshwater with seawater and has a salinity gradient produced by mixing. (2 points).

2. (2 points) Explain how a nutrient pulse can lead to lower overnight oxygen.

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answer: Nutrients can increase production; respiration and decomposition consume oxygen. Nighttime photosynthesis is absent, and mixing also affects the budget. (2 points).

3. (2 points) Identify one watershed action that could reduce nutrient loading and explain its pathway.

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answer: For example, reduce excess fertilizer and retain vegetated buffers to limit nutrient transport in runoff; identify a source and transport pathway. (2 points).

4. (2 points) Explain how traditional ecological knowledge could improve this monitoring plan without replacing measured evidence.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: Collaborate with knowledge holders to identify historical baselines, seasonal changes and appropriate sampling locations; evaluate observations alongside recorded measurements and respect consent/context. (2 points).

5. (2 points) Why should a reef food-web analysis distinguish herbivores from predators?

answer: They affect different trophic pathways; grazing regulates algae while predation affects prey populations, with indirect effects across the web. (2 points).

6. (2 points) Distinguish a harmful algal bloom from an ordinary observation of green water.

answer: A bloom can cause toxins or oxygen stress, but color alone does not establish harmful species, toxins or ecological damage; additional evidence is needed. (2 points).

7. (2 points) A reef fish has fused beak-like teeth and scrapes algae from hard surfaces. Identify its common-name group.

answer: Parrotfish (2 points).

8. (2 points) A starfish has many arms and venomous spines and consumes live coral tissue. Identify it.

answer: Crown of thorns starfish (2 points).

9. (2 points) A reef-building animal secretes a hard calcium carbonate skeleton. Identify its listed group.

answer: Hard/stony coral (2 points).

10. (2 points) A dark urchin has conspicuously long slender spines and grazes algae. Identify its listed group.

answer: Long-spined black sea urchin (2 points).

11. (2 points) A shrimp has red-and-white bands and conspicuous elongated appendages. Identify the listed common-name group.

answer: Banded coral shrimp (2 points).

12. (2 points) An elongate, soft-bodied echinoderm processes sediment on the seafloor. Identify its listed group.

answer: Sea cucumber (2 points).

13. (2 points) Calculate the overnight dissolved-oxygen decline.

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answer: 3 mg/L (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , mg/L.

14. (2 points) Calculate the BOD-bottle oxygen demand under the stated assumptions.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: 3.2 mg/L (2 points).

Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , mg/L.

15. (2 points) A mixture contains equal volumes of freshwater and seawater with salinity 22. Assuming additive volumes and conservative mixing, estimate salinity.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: 11 (2 points).

Numerical target 11; absolute tolerance 0.022. Accepted units: , .

16. (2 points) Aragonite saturation decreases while other factors are held fixed. Explain a relevant concern for stony corals.

answer: Lower aragonite saturation can make calcium-carbonate skeleton formation less favorable and increase dissolution risk; responses depend on organism and conditions. (2 points).

17. (2 points) Explain why high turbidity and high dissolved-solids concentration are not interchangeable observations.

answer: Turbidity reflects light scattering by suspended material. Dissolved substances can change chemistry without producing the same scattering signal. (2 points).

18. (2 points) A field pH value changes between daytime and predawn. Name a biological mechanism consistent with this and one check on the inference.

answer: Photosynthesis/respiration change dissolved CO2 and can shift pH. Check timing, calibration, temperature, depth, mixing and repeated observations before assigning a cause. (2 points).

19. (2 points) A salinometer calibration gives reading 10 at 2% mass/volume salt and reading 30 at 6%. A sample reads 16. Use linear interpolation to estimate concentration.

answer: 3.2 % (2 points).

Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , %.

20. (2 points) How many grams of salt are required for 500 mL of final solution at 3% mass/volume?

answer: 15 g (2 points).

Numerical target 15; absolute tolerance 0.03. Accepted units: , g.